Where Universal Adversarial Perturbations Went Next

Three 2025 papers that cite Moosavi-Dezfooli et al. (CVPR 2017), analysed with one framework

Prepared for the Adversarial ML replication project · Martín · November 2026

The one-sentence takeaway

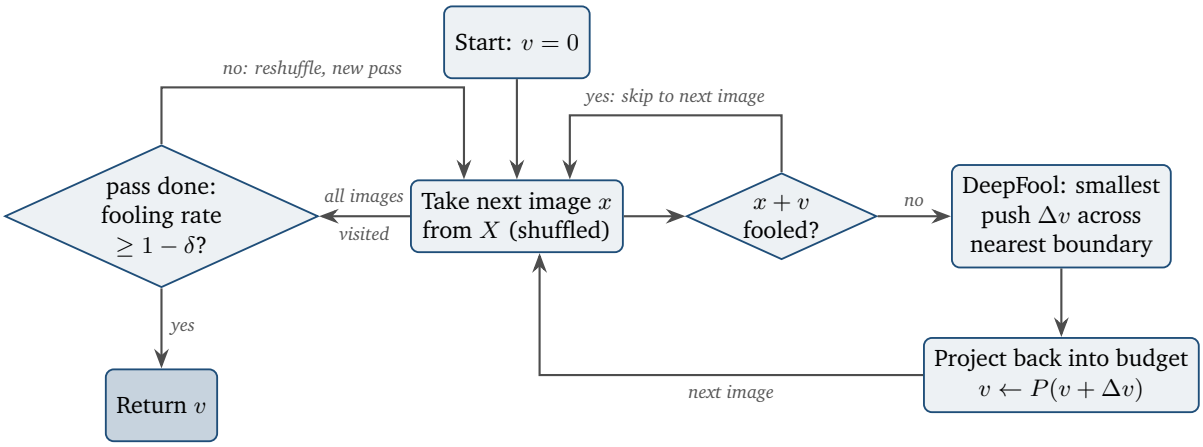
All three papers keep asking the question our paper first raised — *why can one fixed perturbation fool many different inputs?* — and each answers it by finding a **structure shared across inputs**: decision-boundary geometry (original UAP), semantic content inside the perturbation (PSP-UAP), image–text alignment (C-PGC), and a refusal direction inside the model (IRIS).

	PSP-UAP	C-PGC	IRIS
Venue	CVPR 2025	ICCV 2025	NAACL 2025
Domain	Image classifiers (CNNs)	Vision–language models	Large language models
Role of our paper	Blueprint they improve	Baseline that fails when transplanted	Concept exported to text
Replaces Algorithm 1 with	Activation maximisation on crops of the UAP	A generator trained with a contrastive loss	Token search + refusal-direction penalty
Shared structure	Dominant-label semantics	Image–text alignment	Refusal direction
ViT relevance	Admits poor transfer to ViTs	Attacks CLIP-ViT	—

All three papers are peer-reviewed and list the CVPR 2017 UAP paper in their references (checked in the full text). Links are in Section 8.

1 The reference point: our paper in one picture

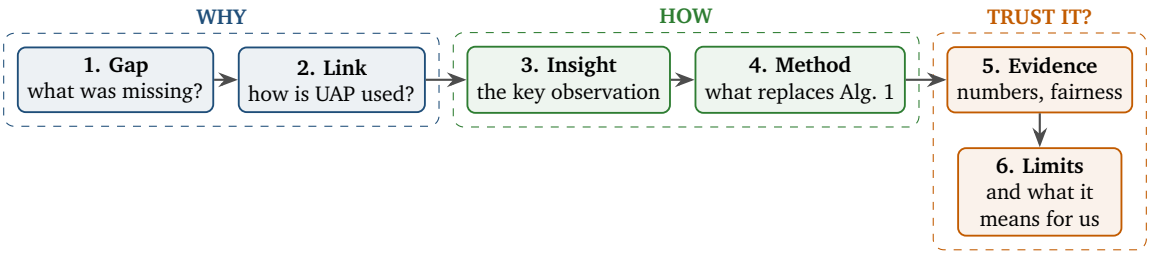
A **universal perturbation** is a single vector v that is added to *any* image and changes the network's prediction most of the time. The paper asks for v that is small, $\|v\|_p \le \xi$, and that fools at least a fraction $1-\delta$ of natural images, measured by the **fooling rate** (prediction with v differs from prediction without it). Algorithm 1 builds v one image at a time:



Why it works. The DeepFool pushes computed for different images point in surprisingly similar directions: the decision boundaries around a dog, a car and a teapot are tilted in similar ways. Each image "votes" for a fooling direction, and because the votes agree, they add up instead of cancelling.

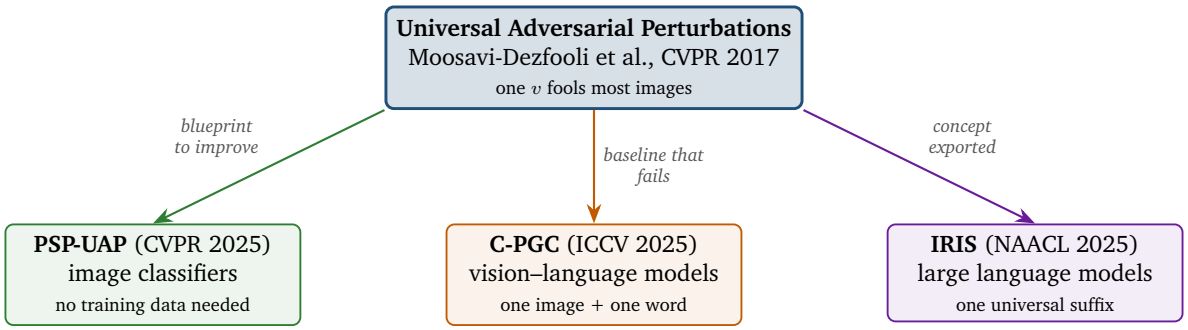
Our replication in numbers (GoogLeNet, ℓ_∞, $\xi = 10/255$, $|X| = 4{,}000$, ImageNetV2): fooling rate 54.6 % on held-out images vs. 78.9 % in the paper ($|X| = 10{,}000$; the paper's own Fig. 6 gives ≈69 % at 4,000 images). The universal v fools 2–3× more images than random noise of the same size, and the most common label it pushes images to is *brain coral* (311 images). Keep these in mind: several of them reappear below.

2 The framework: six questions for any citing paper



1. **The gap.** What problem existed before this paper, and why did the authors care? This is the intuition for why the paper exists.
2. **The link to our paper.** Which role does UAP play: origin of a concept, a baseline to beat, a building block, or an observation they exploit? Which specific idea do they inherit?
3. **The key insight.** The single observation that makes their method work.
4. **The method.** In one or two sentences, and in particular: what replaces DeepFool + projection?
5. **The evidence.** Setup, headline numbers, and whether the comparison is fair.
6. **Limits, and what it means for us.** What the authors admit does not work, and how it connects to our replication and the ViT question.

The family tree



3 Paper 1 — PSP-UAP (CVPR 2025): universal perturbations without data

Chanhui Lee, Yeonghwan Song, Jeany Son (GIST). *Data-free Universal Adversarial Perturbation with Pseudo-semantic Prior*. CVPR 2025. Code: `github.com/ChnanChan/PSP-UAP`.

1. Gap

Our algorithm is *data-dependent*: it needs thousands of real images from the target domain (we used 4,000 ImageNetV2 images). The authors point out that such data is often unavailable to an attacker, which motivates *data-free* UAPs. But existing data-free methods start from random patterns (Gaussian noise, jigsaw images) that carry no semantic content, so the resulting perturbations transfer poorly to models they were not built on.

2. Link to our paper

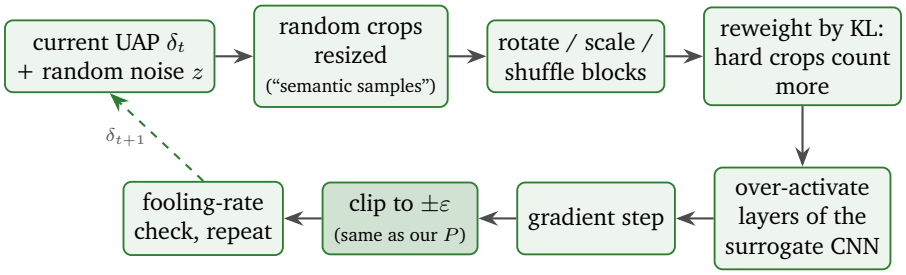
Our paper appears in three ways. (i) **Origin**: they credit it with introducing the image-agnostic attack, and describe it as finding a minimal step with DeepFool at each iteration. (ii) **Metric**: they evaluate with the fooling rate defined there. (iii) **Observation**: their whole method starts from the finding, in our paper and later work, that untargeted UAPs push images toward a few *dominant labels* — so a UAP behaves like a single image with strong semantics.

3. Key insight

If the perturbation looks like an image to the network, it can serve as its own training data. In their Fig. 1, a whole UAP is classified as "safe", while crops of it are classified as "brain coral", "puck" and "slot". **Brain coral is also the top dominant label in our own E6 run** (311 fooled images) — a related effect you can show directly from our notebook.

4. Method — what replaces Algorithm 1

DeepFool is gone. The perturbation is trained by gradient ascent on a loss that over-activates the network's internal layers, computed on random crops of (UAP + noise), with random transformations and a reweighting that favours crops the UAP does not fool yet. The clipping step to $\pm\varepsilon$ is the same convex projection as our `proj_lp`.



5. Evidence

Same budget as our notebook: ℓ_∞, $\varepsilon = 10/255$. On five ImageNet CNNs (AlexNet, VGG16, VGG19, ResNet152, GoogLeNet), PSP-UAP has the best average white-box fooling rate among data-free methods (89.95 %), with a 12.5-point gain on ResNet152. In black-box transfer it even beats data-dependent methods that use real training images. **Caution:** they evaluate on the original ImageNet validation set, not ImageNetV2, so their numbers are not directly comparable to our 54.6 %.

6. Limits — and what it means for us

The most relevant point for our ViT question: the authors state their method transfers poorly to ViT-based models, and that this limitation appears to be shared by both data-free and data-dependent UAP methods. Used as a surrogate, DeiT3 (a ViT) gave weak results, which they attribute to the methods being designed for CNNs. **Our E7 experiment probes exactly this gap with the original Algorithm 1.**

4 Paper 2 — C-PGC (ICCV 2025): one perturbation for vision–language models

Hao Fang, Jiawei Kong, Wenbo Yu, Bin Chen, Jiawei Li, Hao Wu, Shu-Tao Xia, Ke Xu (Tsinghua / HIT Shenzhen). *One Perturbation is Enough: On Generating Universal Adversarial Perturbations against Vision-Language Pre-training Models*. ICCV 2025, pp. 4090–4100.

1. Gap

Vision–language pre-training (VLP) models such as CLIP, ALBEF and BLIP were already known to be attackable, but every existing attack was *instance-specific*: a new perturbation for each image–text pair, which is expensive. The question is whether a single universal perturbation can fool these models across many pairs.

2. Link to our paper — the baseline that fails

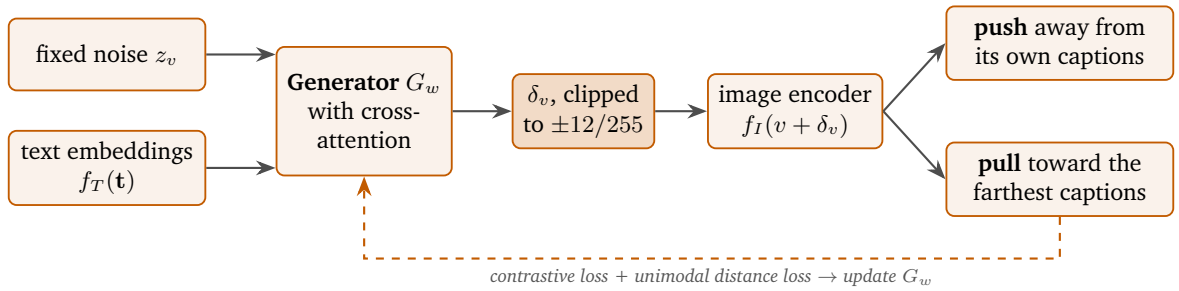
This is the most interesting role our paper plays. The authors *transplant* the original UAP (DeepFool-based, plus a PGD variant for fairness) and a generative method (GAP) to VLP models, optimising the distance between the perturbed image's embedding and its matching text. Their Fig. 2 shows low attack success, especially in black-box transfer. Their diagnosis: these methods look only at the image and ignore the text modality and the cross-modal information VLP models rely on. They also observe that the generator-based GAP beats the UAP methods.

3. Key insight

VLP models work because *contrastive learning* aligns matching images and texts. So an effective universal attack should **break that alignment**, rather than just change one output — and the natural tool to break it is contrastive learning itself, turned against the model.

4. Method — what replaces Algorithm 1

A neural network (the generator) is trained to output the perturbation. It turns a fixed noise input into an image perturbation, using cross-attention on the text embeddings as extra guidance. The training loss pushes the adversarial image away from its own captions and pulls it toward the captions most distant from it, plus a term pushing it away from the clean image in feature space. A symmetric branch produces a one-word text perturbation.



5. Evidence

Budget: $\ell_\infty \leq 12/255$ for the image, one substituted word for the text. Six VLP models (ALBEF, TCL, X-VLM, CLIP-ViT, CLIP-CNN, BLIP) and four tasks (retrieval, captioning, visual grounding, visual entailment). A single pair of perturbations reaches an average white-box attack success rate of nearly 90 % on two large datasets; black-box transfer is also strong, e.g. 72.51 % from ALBEF to CLIP-CNN on image retrieval. Ablations show the contrastive loss is the main contributor, especially for transfer.

6. Limits — and what it means for us

The image and text perturbations are optimised separately, without modelling how they interact. For our project, the pattern to remember: **when the target changes from "flip a label" to "break an alignment", the objective has to change, not just the model**. It is also a sign of a broader shift from per-image optimisation (DeepFool) to learned generators.

5 Paper 3 — IRIS (NAACL 2025): universal attacks on language models

David Huang, Avidan Shah, Alexandre Araujo, David Wagner, Chawin Sitawarin (UC Berkeley / NYU). *Stronger Universal and Transferable Attacks by Suppressing Refusals*. NAACL 2025, pp. 5850–5876. Safety red-teaming research: we only use its ideas, not its examples.

1. Gap

Optimisation-based jailbreaks such as GCG search for an adversarial suffix that makes an aligned LLM comply with a harmful request. The common belief was that these suffixes are sample-specific, and that making them universal and transferable requires optimising over many requests and many models at once — expensive, and still weak against frontier models.

2. Link to our paper — the concept, exported

In their background section they define the *universal attack*, citing our paper: a perturbation created for one sample that also works on many unseen samples. Together with transfer attacks from the image literature, this gives them the two realistic threat models they study for LLMs.

3. Key insight

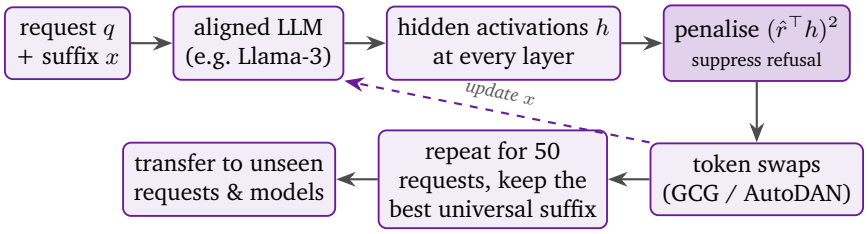
A small fraction (under 10 %) of suffixes optimised on *one* request and *one* model are already universal and transferable. This reverses what adversarial-example research would predict: the best universal suffix beats the average universal one, which beats the individually optimised ones. Their explanation: these suffixes switch off the model's **general refusal mechanism** rather than forcing one specific answer.

4. Method — what replaces Algorithm 1

IRIS adds one term to existing optimisers (GCG, AutoDAN): besides making a compliant answer likely, it penalises the component of the model's hidden activations along a pre-computed "refusal direction" $\hat{r}$ (the difference between average activations on harmful and harmless prompts):

$$\mathcal{L}_{\text{IRIS}}(x) = -(1-\beta)\,\log p_\theta(y \mid q \,\|\, x) \;+\; \beta \sum_h \left(\hat{r}^\top h\right)^2.$$

The suffix x plays the role of our v; discrete token search replaces DeepFool.



5. Evidence

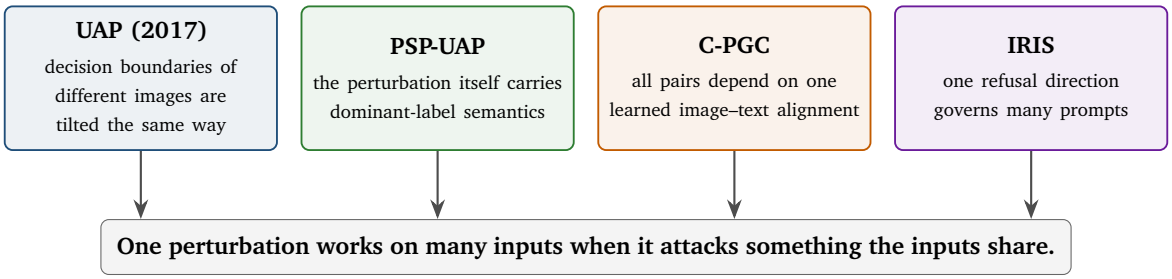
Suffixes are optimised on Llama-3 and never on the target models. Against the Circuit Breaker defense, IRIS reaches 25 % success versus 2.5 % for white-box GCG. It transfers to GPT-4o (76 %), GPT-4o-mini (86 %), o1-mini (54 %), o3-mini (66 %) and deepseek-reasoner (90 %). The "best universal" effect alone lifts white-box GCG on Llama-2 from 34 % to 88 %.

6. Limits — and what it means for us

Success is judged by an automated LLM judge, which the authors say undercounts successes, so the rates are lower bounds; results also vary a lot with the random seed and the source request. **Connection to our paper (our interpretation, not the authors' claim):** UAP explains universality through shared geometry — one direction crosses many decision boundaries. IRIS points to a similar mechanism in LLMs: one shared internal direction (refusal) governs behaviour across many prompts, so one suffix that suppresses it works on many prompts.

6 Putting it together

The common thread: a shared structure across inputs



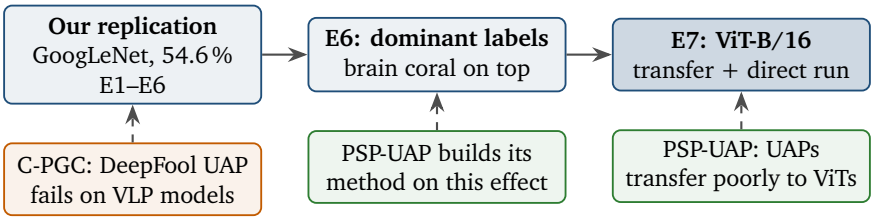
Side-by-side comparison

	PSP-UAP	C-PGC	IRIS
What it perturbs	Image pixels	Image pixels + one word	Text suffix (tokens)
Budget	$\ell_\infty \leq 10/255$	$\ell_\infty \leq 12/255$, 1 word	20-token suffix
Success measure	Fooling rate (from our paper)	Attack success rate on V+L tasks	Jailbreak success (LLM judge)
Needs real data?	No (data-free)	Yes (image–text pairs)	50 harmful requests
Optimiser	Gradient ascent on layer activations	Trained generator	Discrete token search
Headline result	Best data-free fooling rate on 5 CNNs	≈90 % white-box with one pair	76 % transfer to GPT-4o
Admitted limit	Weak on ViTs	No image–text interaction	Judge undercounts; high variance

Three trends since 2017

1. **DeepFool has been replaced.** None of the three keeps Algorithm 1's inner step: they use direct gradient optimisation, learned generators, or discrete search. The clip-to-budget projection survives wherever the perturbation is continuous.
2. **The target moved beyond CNN classifiers.** From image labels to cross-modal alignment to LLM safety behaviour — and ViTs remain a weak spot for UAP methods built on CNN intuitions.
3. **The "why" became the method.** Our paper *explained* universality with shared geometry after the fact; the newer papers *start* from a hypothesis about the shared structure and build the objective around it.

7 How this connects to our notebook



Talking points for the review

- **What we did:** replicated Algorithm 1 in PyTorch; 54.6 % fooling on held-out ImageNetV2, with the gap to the paper explained by fewer images (4,000 vs. 10,000) and the ℓ_∞ clipping plateau.
- **How we followed the paper forward:** Google Scholar "cited by", three peer-reviewed 2025 papers across images, vision–language and LLMs.
- **What they tell us:** DeepFool has been replaced, the targets have moved beyond CNNs, and UAPs remain weak on ViTs — which our E7 experiment tests with the original algorithm.
- **One sentence:** one perturbation works on many inputs when it attacks a structure those inputs share.

8 References

[1] S.-M. Moosavi-Dezfooli, A. Fawzi, O. Fawzi, P. Frossard. *Universal adversarial perturbations*. CVPR 2017, pp. 1765–1773.

[2] C. Lee, Y. Song, J. Son. *Data-free Universal Adversarial Perturbation with Pseudo-semantic Prior*. CVPR 2025. https://openaccess.thecvf.com/content/CVPR2025/papers/Lee_Data-free_Universal_Adversarial_Perturbation_with_Pseudo-semantic_Prior_CVPR_2025_paper.pdf — arXiv:2502.21048.

[3] H. Fang, J. Kong, W. Yu, B. Chen, J. Li, H. Wu, S.-T. Xia, K. Xu. *One Perturbation is Enough: On Generating Universal Adversarial Perturbations against Vision-Language Pre-training Models*. ICCV 2025, pp. 4090–4100. https://openaccess.thecvf.com/content/ICCV2025/html/Fang_One_Perturbation_is_Enough_On_Generating_Universal_Adversarial_Perturbations_against_ICCV_2025_paper.html — arXiv:2406.05491.

[4] D. Huang, A. Shah, A. Araujo, D. Wagner, C. Sitawarin. *Stronger Universal and Transferable Attacks by Suppressing Refusals*. NAACL 2025, pp. 5850–5876. https://aclanthology.org/2025.naacl-long.302.pdf

[5] P. Benz, S. Ham, C. Zhang, A. Karjauv, I. S. Kweon. *Adversarial Robustness Comparison of Vision Transformer and MLP-Mixer to CNNs*. arXiv:2110.02797, 2021.